

Smart Breadboard Diagnostic Trainer

How the system works, start to finish. No electronics or programming knowledge needed.

1 What it does

Electronics students build circuits on a **breadboard**, a plastic board full of holes. When a circuit does not work, the board gives no clue why. The student waits for a teacher who has forty other students. The teacher points at the wrong wire and walks away. Nobody learns how to find the mistake.

We put that board in a web browser. The student builds the circuit in 3D. The system compares it with the circuit the teacher already built, finds the mistake, points at the exact row, and asks a question instead of giving the answer.

2 Two modes

Almost every decision in this project comes from this one difference.

	FREE BUILD	TEACHER'S TASK
What it is	Practice. Build anything.	An assignment from a teacher.
Graded?	No	Yes, automatically
Deadline or timer?	No	Whatever the teacher set
Help while building	Full	Only what the teacher allowed
AI feedback	Straight away	Only after submitting

Why the AI stays quiet during a graded task

In Free Build there is nothing to cheat on, so the AI helps right away. In a graded task the student is being marked. If the AI explains the mistake while they can still fix it, we are grading work our own software corrected. So the feedback is written when they submit, then held until the deadline.

3 How the AI fits in

Think of a blood test. A machine runs the test and gives numbers. A doctor explains those numbers in words you understand. The doctor does not re-run the test by guessing.

Our system works the same way. A fixed **checker** decides what is right and wrong. The **AI Coach** only puts that result into friendly words.

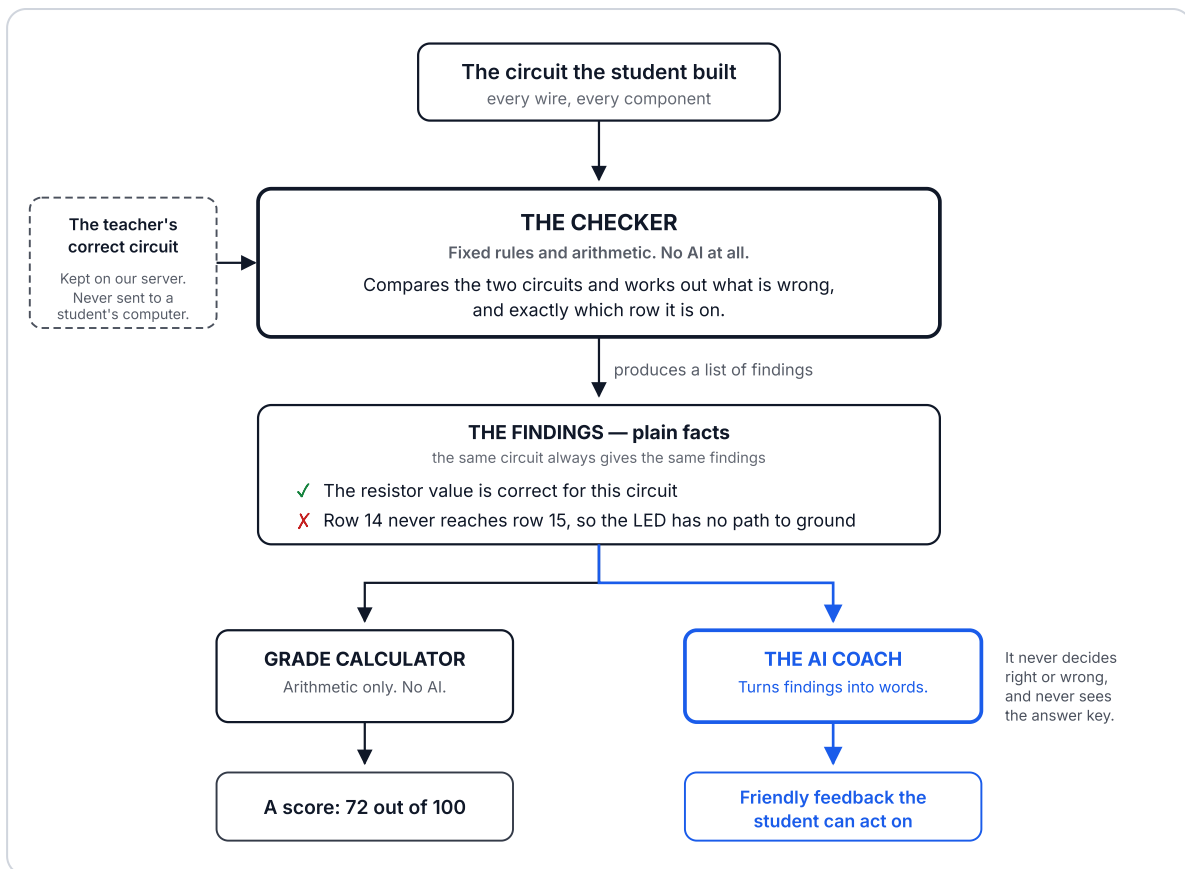


Figure 1 — the AI comes last. Everything that decides right, wrong, and the score happens before it. Turn the AI off and students still get a correct diagnosis and a correct grade, just in blunter words.

Three sentences to remember

1. The checker decides what is wrong.
2. The AI only puts it into words.
3. The AI may never mention a problem the checker did not find.

4 When the student may read it

Same checker, same AI. Only the timing changes.

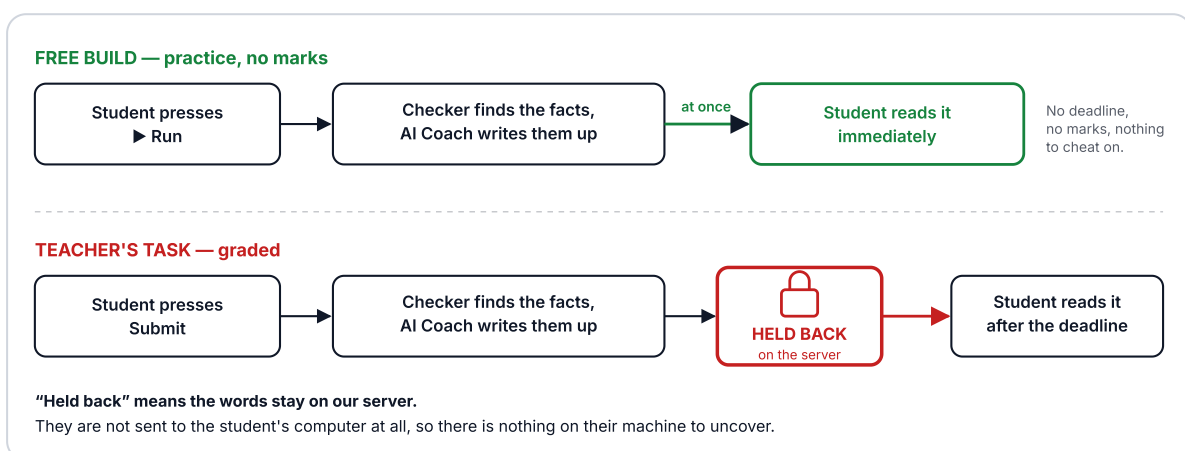


Figure 2 — one pipeline, two release times. Hiding text inside the browser is not a lock. The words stay on the server, and the server decides when the deadline has passed.

5 Who does what

Teacher

1. Create a class and share the join code.
2. Write an exercise: title, goal, parts list.
3. **Build the correct circuit once.** The system saves it as the answer key. Everything else depends on this step.
4. Assign it to a class as a task, with a deadline and any other settings.
5. Grades appear automatically as students submit.
6. Feedback opens at the deadline. Override any score if needed.
7. Read the class report and export it.

Student

1. Register and join a class with the code.
2. See what is due, with a countdown.
3. Read the rules before starting: time limit, attempts, help allowed.
4. Build the circuit on the 3D board.
5. Submit, or the timer submits for them.
6. See the score straight away, with a breakdown.
7. Read the AI feedback when it opens.
8. Practise any time in Free Build.

6 What a teacher can set on a task

SETTING	OPTIONS	DEFAULT
Deadline	When it opens, when it is due, when it closes for good	No deadline
Late work	Blocked, accepted, or accepted with a penalty per day	Blocked
Timer	None; soft, which warns but lets them continue; hard, which submits at zero	None
Attempts	How many, and whether to keep the highest, latest, first, or average	Unlimited, highest
Help allowed	None for exams; safety warnings only; or full, with a Check my work button	Safety warnings
Hints	Allowed or not, capped, and optionally worth a point deduction	Allowed and free
Grading	Automatic, automatic with review, or manual. Total points and pass mark	Automatic, 100, 70%
Feedback opens	Immediately, on submit, at the deadline, or when the teacher releases it	At the deadline

Two things the system enforces on its own

Deadlines and timers run on our server, so changing the clock on a laptop does nothing. And immediate feedback cannot be combined with automatic grading and multiple attempts, because that lets a student submit, read the fix, and resubmit until they score full marks.

7 How the grade is worked out

The teacher already built the correct circuit, so the system knows the answer before anyone starts. No human marking is needed.

MARKING LINE	POINTS	WHAT EARNS THEM
Circuit correctness	60	How close the connections are to the teacher's, with partial credit
Component selection	10	Right parts, right values
Polarity and orientation	10	Nothing inserted backwards
Nothing unsafe	10	No short circuit when they submit
Process	10	Hints used, and problems solved without help

The score and its breakdown appear the moment a student submits. Only the AI feedback waits. A number on its own is not a correction, so showing it early costs nothing.

The first thing to test when we demo this

A circuit that is **correct but built on different rows** must score full marks. Students do not all build in the same place. If that ever fails, nothing else about the grading matters.

8 The four rules

1. The AI never decides whether a circuit is right, and never gives a grade.
2. The AI may never mention a problem the checker did not find.
3. On a graded task, nothing reaches the student before the deadline, and the holding happens on the server.
4. The teacher's answer key never leaves the server.

If the AI is unavailable, or writes something it should not have, students still get the checker's diagnosis in our own wording. It reads more plainly. It is never absent and never wrong.